

Zhuowei Chen

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EDUCATION

Guangdong University of Foreign Studies (GDUFS)

B.E. in Software Engineering. Advisor: Lianxi Wang

GPA: 3.80/4.00

Guangzhou, China

Sept 2021 - June 2025

University of California, Berkeley (UCB)

Courses: NLP, Introduction to AI, Computer Security

GPA: 4.00/4.00

Berkeley, CA

Aug 2023 - Jan 2024

PUBLICATIONS

* represents equal contributions and † represents the corresponding author.

1. **Zhuowei Chen**, Yuben Wu, Xinfeng Liao, Yujia Tian, Lianxi Wang[†].
[An Effective Deployment of Diffusion LM for Data Augmentation in Low-Resource Sentiment Classification.](#)
The 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024.
2. Lianxi Wang, Yujia Tian*, **Zhuowei Chen***[†].
[Enhancing Hindi Feature Representation Through Fusion of Dual-Script Word Embeddings.](#)
The Joint Conference on Computational Linguistics, Language Resources and Evaluation, LREC-COLING 2024.
3. **Zhuowei Chen**, Yujia Tian, Lianxi Wang[†], Shengyi Jiang.
[A Distantly-Supervised Relation Extraction Method Based on Selective Gate and Noise Correction.](#)
The 22nd China National Conference on Computational Linguistics, CCL 2023.
4. **Zhuowei Chen**, Qiannan Zhang, Shichao Pei.
[Injecting Universal Jailbreak Backdoors to LLMs in Minutes.](#)
(Under review @ ICLR 2025, **Avg. Rating 6.67**)
5. Lianxi Wang, Huayu Huang, **Zhuowei Chen**[†].
A Knowledge-Augmented and Label-Aware Framework for Multi-Label Text Classification.
(Under review @ Journal of Computer Science and Technology)
6. Lianxi Wang, Yujia Tian*, **Zhuowei Chen***, Mutong Li, Nankai Lin[†].
EditMDS: An Iterative Optimization Method for Multi-Document Summarization Based on Edit Operations.
(Under review @ ICASSP)

SELECTED HONORS

- **China National Scholarship** (Top 0.2%) Ministry of Education of the PRC, 2024
- **First-class Scholarship** (Top 4%) Guangdong University of Foreign Studies, 2024
- **First-class Scholarship** (Top 4%) Guangdong University of Foreign Studies, 2023
- **Silver Medal** (Top 5%) National College Computer Design Competition, 2022
- **Bronze Medal** China Undergraduate Mathematical Contest in Modeling (Regional), 2023

RESEARCH EXPERIENCE

University of Massachusetts Boston

Research Intern

Supervisor: Dr. Shichao Pei

Boston, MA

March 2024 - Oct 2024

- JailbreakLLM: Exploring Novel Jailbreak Backdoor Attacks on LLMs. (ICLR 2025)
 - Proposed a novel method to inject universal backdoors into LLMs without additional datasets or extensive computational overhead.
 - Developed a multi-node target estimation strategy that preserves attack stealthiness, effectively overwhelming and disabling the internal safety mechanisms of large language models.
 - Executed comprehensive experiments, confirming a high jailbreak success rate and highlighting the urgency for advanced defensive strategies in LLMs.

Guangzhou Key Laboratory of Multilingual Intelligent Processing

Undergraduate Research Student

Supervisor: Prof. Lianxi Wang

Guangzhou, China

Nov 2021 – March 2024

- Deploying Diffusion LM for Data Augmentation in Text Classification. (EMNLP 2024)
 - Fine-tuned LMs with a diffusion objective to capture in-domain knowledge and generate samples by reconstructing label-related tokens.
 - Designed attention-based mask schedule for the diffusion LM, balancing domain consistency, label consistency, and context diversity.
 - Conducted analyses and visualizations to study its underlying mechanism, followed by experiments validating its effectiveness across various low-resource scenarios.
- Enhancing Hindi Representations via Fusion of Pre-trained Language Models. (COLING 2024)
 - Proposed a method to enhance Hindi feature representation by combining Devanagari and Romanized Hindi pre-trained language models.
 - Conducted an in-depth comparison of different feature fusion techniques, including concatenation, summation, and cross-attention.
 - Ablations and extensive NLU task experiments show the superiority of our method, demonstrating the potential of multi-script integration to enhance low-resource language models.
- Distantly Supervised Relation Extraction (DSRE) with Learning-with-Noise Methods. (CCL 2023)
 - Combined selective gate and noise correction training framework for DSRE, which performs data selection and corrects noise labels during a three-stage training process.
 - Experiments demonstrated state-of-the-art performance, revealing a promising new approach for applying training-with-noise techniques in NLP.
- Multi-Label Text Classification (MLTC) with Knowledge Augmentation and Span Prediction.
 - Integrated span-prediction with an adapted GNN-based knowledge augmentation module to enhance MLTC.
 - Conducted visualizations and analyses to study its working mechanism, emphasizing the critical role of incorporating domain-specific knowledge for LM.

SELECTED PROJECTS

- BiasLLM: Adversarial Knowledge Editing Attacks on LLMs.
 - Combined GNNs with locate-then-edit techniques (ROME) to attack Llama-2, successfully exposing significant biases within the model.
- Multimodal NLP: Image-Text Interfacing with CLIP and Rational Speech Acts.
 - Used the CLIP model for image and caption retrieval, and further improved retrieval effectiveness by developing and applying a Rational Speech Acts inference procedure.

WORK EXPERIENCE

AI Lab, Wisers Information Ltd.

NLP Research Intern

Hong Kong, China

Dec 2023 - March 2024

- Built BERT-based textual classification models with human-annotated social media content.
- Applied transformers for time series regression to predict regional arrivals.

LANGUAGES & SKILLS

- **Programming:** Python, Java, SQL, JS/HTML/CSS, C/C++, Golang.
- **Languages:** English(IELTS 7.5), Mandarin(Native), Cantonese(Native).

OTHER RELATED EXPERIENCE

- **Conference Attendance.** Poster and oral presentation on *EMNLP 2024* and *LREC-COLING 2024*.
- **Teaching Assistance.** TA for Language Processing Technique.
- **Member of Publicity at Student Union.** Organized poster presentations promoting AI equity.
- **Volunteer Lecturer at Dongguan Library.** Introduced basics of AI to the public.

陈卓玮

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教育

广东外语外贸大学 (GDUFS)
软件工程学士学位; 导师: 王连喜
GPA: 3.80/4.00

中国, 广州
2021.09 - 2025.06

加州大学伯克利分校 (UC Berkeley)
课程: NLP, AI 导论, 计算机安全
GPA: 4.00/4.00

美国, 伯克利
2023.08 - 2024.06

相关发表

* 代表同等贡献, † 代表通讯作者。

- Zhuowei Chen**, Yuben Wu, Xinfeng Liao, Yujia Tian, Lianxi Wang[†]. **An Effective Deployment of Diffusion LM for Data Augmentation in Low-Resource Sentiment Classification**. *The 2024 Conference on Empirical Methods in Natural Language Processing, EMNLP 2024*.
- Lianxi Wang, Yujia Tian*, **Zhuowei Chen***[†]. **Enhancing Hindi Feature Representation Through Fusion of Dual-Script Word Embeddings**. *The Joint Conference on Computational Linguistics, Language Resources and Evaluation, LREC-COLING 2024*.
- Zhuowei Chen**, Yujia Tian, Lianxi Wang[†], Shengyi Jiang. **A Distantly-Supervised Relation Extraction Method Based on Selective Gate and Noise Correction**. *The 22nd China National Conference on Computational Linguistics, CCL 2023*.
- Zhuowei Chen**, Qiannan Zhang, Shichao Pei. **Injecting Universal Jailbreak Backdoors to LLMs in Minutes**. (在审 @ ICLR 2025, 均分 6.67)
- Lianxi Wang, Huayu Huang, **Zhuowei Chen**[†]. A Knowledge-Augmented and Label-Aware Framework for Multi-Label Text Classification. (在审 @ Journal of Computer Science and Technology)
- Lianxi Wang, Yujia Tian*, **Zhuowei Chen***, Mutong Li, Nankai Lin[†]. EditMDS: An Iterative Optimization Method for Multi-Document Summarization Based on Edit Operations. (在审)

研究及工作经历

麻省大学 (波士顿)
研究实习生
导师: 裴仕超

美国, 波士顿
2024 年 3 月-2024 年 12 月

- JailbreakLLM: 探索对大型语言模型 (LLMs) 的新型后门攻击。(ICLR 2025)
 - 提出了一种无需额外数据集或大量计算开销的方法, 向 LLM 中注入通用后门。
 - 开发了多节点目标估计, 在保持攻击隐匿性同时, 有效削弱并绕过 LLM 的内部安全机制。
 - 进行了全面实验, 验证了高成功率的越狱效果, 强调了开发高级防御策略的紧迫性。

慧科讯业 | 人工智能实验室
NLP 实习生

中国, 深圳
2023 年 12 月-2024 年 3 月

- 基于 BERT 的文本分类模型构建, 使用人工标注的社交媒体内容。
- 应用 transformer 模型进行时间序列回归, 以预测到港访客量。

广州多语种智能处理重点实验室
本科生
导师: 王连喜

中国, 广州
2021 年 11 月-2024 年 3 月

- 使用扩散语言模型 (Diffusion LM) 进行文本分类的数据增强。(EMNLP 2024)
 - 通过扩散目标对语言模型进行微调以捕获领域知识, 并通过重构标签相关的标记生成样本。
 - 设计了基于注意力的掩码调度机制, 以平衡领域一致性、标签一致性和上下文多样性。
 - 通过分析和可视化研究其工作机制, 实验验证了其在低资源场景中的有效性。

- 通过融合预训练语言模型增强印地语表示能力。(COLING 2024)
 - 提出一种结合天城文和罗马化印地语预训练语言模型的方法，以增强印地语特征表示。
 - 深入比较了不同特征融合技术，包括拼接、求和和交叉注意力。
 - 消融实验和充分的 NLP 实验表明，该方法在增强低资源语言模型方面具有优势。
- 使用噪声学习方法进行远程监督关系抽取。(CCL 2023)
 - 结合选择门机制与噪声校正训练框架，通过三阶段训练框架进行数据选择和噪声标签校正。
 - 实验表明，该方法在性能上达到了 SOTA 水平，为在自然语言处理中应用噪声学习方法提供了新的可能性。
- 结合知识增强和跨度预测的多标签文本分类。
 - 结合跨度预测和基于图神经网络的知识增强模块，以改进多标签文本分类。
 - 进行了可视化和分析以研究其工作机制，强调了整合领域知识对于语言模型的重要性。

项目经历

- BiasLLM：对 LLM 的对抗性知识编辑攻击。
 - 将图神经网络（GNN）与“定位然后编辑”（ROME）技术相结合，对 Llama-2 模型进行攻击，成功揭示了模型中的显著偏差。
- 多模态自然语言处理：使用 CLIP 和理性语言行为理论实现图文交互。
 - 使用 CLIP 模型进行图像和标题的检索，并通过开发理性语言行为推断程序进一步提升检索效果。

语言与技能

- 编程语言: Python, SQL, Java, JS/HTML/CSS, C/C++, Golang
- 语言能力: 英语 (IELTS 7.5)

荣誉

- 中国国家奖学金教育部, 2024
- 综合一等奖学金(Top 4%) 广东外语外贸大学, 2024
- 综合一等奖学金(Top 4%) 广东外语外贸大学, 2023
- 二等奖(Top 5%) 中国计算机设计大赛, 2022
- 三等奖全国大学生数学建模竞赛 (广东省), 2023